EXACT COUNTERMODEL SPECTRA FOR PERIODIC RIGHT TRANSLATIONS IN FINITE MAGMAS

PEDRO LIRON DE ROBLES

Abstract. We determine the finite countermodel sizes of two families of equational implications in magmas with periodic right translations. For each positive return exponent k, every implication in either family has the same spectrum. There are no countermodels for $k = 1$. Otherwise the minimum size is 3 if 6 divides k, 4 if exactly one of 2 and 3 divides k, and the least prime divisor of k if k is coprime to 6. Every size at least the minimum occurs. The lower bound reduces to orbit lengths and obstructions on at most three elements. For every divisor $d \geq 5$ of k, two conjugate d-cycles give a common witness on d elements. The spectrum theorem is verified in Lean 4.

1. The spectrum theorem

Let M be a nonempty magma, with product xy and right translations $R_y(x) = xy$. For a positive integer k, consider the full and diagonal return laws

$$P_k\colon R_y^k(x) = x \quad \text{and} \quad D_k\colon R_x^k(x) = x,$$

where powers denote iteration and all variables in an identity are universally quantified. The law P_k makes each R_y a permutation whose order divides k. The law D_k only imposes a return condition at the index of each translation.

The following are the side identities B_i and the conclusions C_j.

B_1	$x(xx) = x(xy)$	C_1	$x(xx) = xx$
B_2	$x(xy) = x(xz)$	C_2	$x(xy) = xx$
B_3	$x(yx) = x(yy)$	C_3	$x(yx) = xy$
B_4	$x(yx) = x(yz)$	C_4	$x(yy) = xy$
B_5	$x(yy) = x(yz)$	C_5	$x(yz) = xy.$

A countermodel to a pair of premises and a conclusion satisfies both premises and fails the conclusion at some assignment. The *countermodel spectrum* is the set of positive sizes of such magmas.

Theorem 1.1. *Fix $k \geq 1$. The countermodel spectrum of each of the implications*

$$P_k, B_i \implies C_j \qquad (1 \leq i, j \leq 5),$$
$$D_k, B_i \implies C_j \qquad (i \in \{4, 5\},\ j \in \{1, 2\})$$

consists of exactly the positive integers n satisfying at least one of the conditions

(1) $3 \leq n$ *and* $6 \mid k$,

(2) $4 \leq n$ *and* $(2 \mid k$ *or* $3 \mid k)$,

(3) *there exists* $d \mid k$ *with* $5 \leq d \leq n$.

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Corollary 1.2. *For each implication of Theorem 1.1, the spectrum is empty when $k = 1$. For $k > 1$ it consists of all integers at least $m(k)$, where*

$$m(k) = \begin{cases} 3 & \text{if } 6 \mid k, \\ 4 & \text{if exactly one of } 2 \text{ and } 3 \text{ divides } k, \\ \text{the least prime divisor of } k & \text{if } \gcd(k, 6) = 1. \end{cases}$$

In particular, $m(p) = p$ for every prime $p \geq 5$, so the minimum sizes are unbounded as k varies.

Proof. If $k = 1$, all three conditions fail. If $6 \mid k$, the first condition applies from size 3 onward. If exactly one of 2 and 3 divides k, the second applies from size 4 onward. In the remaining case $k > 1$, every divisor of k other than 1 is at least 5. The least such divisor is prime, and condition (3) applies from that size onward. □

The exponent-two case has threshold four. The drop to three at multiples of six comes from mixing a transposition with a three-cycle. The constructions in Section 3 show that a single table can witness every implication in the theorem at each admissible pair (k, n).

2. Reductions and lower bounds

Write $u \sim v$ when $R_u = R_v$. Then B_5 says that $yz \sim yy$ for every y, z, and C_5 says that $yz \sim y$ for every y, z. Equivalently, B_5 says that the $\sim$-class of $R_z(y)$ is independent of z. Substitutions give $B_1 \Leftrightarrow B_2$ and $B_4 \Leftrightarrow B_5$. For the latter equivalence, use $z = y$ in B_4 and compare the outputs with $z = x$ and with arbitrary z in B_5. The identity B_5 implies B_1 by setting $y = x$, and B_3 by setting $z = x$. Thus every side identity implies at least one of B_1 and B_3, and B_5 implies all five. Every C_j follows from C_5 by substitution, and under B_5 we have $C_1 \Leftrightarrow C_2$ because $x(xy) = x(xx)$. Conversely, a single assignment with $x(xx) \neq xx$ refutes every C_j by setting all its variables equal to x.

Lemma 2.1. *On a magma of size at most three, P_2 together with either B_1 or B_3 implies C_5. On a magma of size at most three, D_2 together with B_5 implies C_1.*

Proof. First assume P_2. Each column is the identity or a transposition. Suppose two distinct transpositions occur. The size is then three, and we may write the two transpositions as $P = (a\,b)$ and $Q = (a\,c)$.

Under B_1, the map $F_x(t) = xt$ has constant square for every x. Since b, c are both in the image of F_a, it follows that $R_b(a) = R_c(a)$. At least one of R_b, R_c is P or Q. An involution on three points is determined by the image of a point it moves, so $R_b = R_c$. Up to exchanging b and c, the columns are $R_a = P$ and $R_b = R_c = Q$. But then $F_b(a) = a$ and $F_b(b) = b$, contrary to the constancy of F_b^2.

Under B_3, if every column is a transposition, distinct columns differ at every point. The equality $R_{yx}(x) = R_{yy}(x)$ thus gives $R_{yx} = R_{yy}$ for all x, y. This is B_5, which implies B_1, so the preceding argument applies. If an identity column occurs, the three columns must be id, P, Q, each occurring once. Their values at a are distinct, so B_3 at $x = a$ gives $ya = yy$ for every y. Hence R_a must be the identity. Indeed, if R_a were one of P, Q, evaluating at the index of the other would equate two distinct transpositions at one point. Now $R_y(y) = y$ for every y, so $R_b = Q$ and $R_c = P$. But B_3 at $x = b, y = c$ equates $b(cb) = b$ with $b(cc) = a$.

Thus at most one transposition occurs. If all columns are equal, C_5 holds. Otherwise the columns take the two values id and $T = (u\,v)$. If $R_u \neq R_v$, choose the names so that $R_u = id$ and $R_v = T$. Then F_u fixes both u and v, contrary to B_1. For B_3, taking $x = u, y = v$ gives

$u(vu) = v$ and $u(vv) = u$, also a contradiction. Therefore $R_u = R_v$. Each translation now preserves every $\sim$-class, which gives $yz \sim y$ and hence C_5.

Next assume D_2 and B_5. If C_1 fails at a, let $b = aa$, $c = ab$ and $d = bb$, so $b \neq c$. We have $c \sim b$ by B_5, and $ba = a$ by D_2 at a, so $a \sim d$ by B_5. If $a \sim b$, then $ab = aa$ and $c = b$. Thus the classes of a, d and b, c are distinct. Finally $a \neq d$, since $a = d$ and D_2 at b would give $ab = (bb)b = b$. The four elements a, b, c, d are distinct. □

Lemma 2.2. *On a magma of size at most three, P_3 together with either B_1 or B_3 implies C_5. On a magma of size at most three, D_3 together with B_5 implies C_1.*

Proof. Under P_3, all columns are the identity when the size is less than three. On three points the possible columns are the identity and the two three-cycles. Any distinct pair differs at every point. Thus B_1 gives $R_{xy} = R_{xx}$ for all x, y, and B_3 gives $R_{yx} = R_{yy}$ for all x, y. Either identity yields B_5. If $R_e = \mathrm{id}$ for some e, then B_5 gives $R_{yz} = R_{ye} = R_y$, yielding C_5. If both three-cycles P and P^{-1} occur, B_5 gives $P(y) \sim P^{-1}(y)$ for every y. Since P^2 is transitive, this would make all columns equal. The remaining possibility is a single distinct column, which also yields C_5.

Now assume D_3 and B_5, and suppose C_1 fails at a. Then $R_a(a) \neq a$, so the return orbit of a has length three. The size is three and $P = R_a$ is a three-cycle. If all columns are equal then $R_{aa} = R_a$ and $a(aa) = aa$. If all columns are distinct, B_5 gives $yz = yy$ for all y, z, which would make all columns equal. There are therefore exactly two distinct columns P, Q.

Let s be the unique index in the singleton $\sim$-class and let $t = P^{-1}(s)$. Since $P(t) = s$ and $P(s) \neq s$, the class condition B_5 gives $Q(t) = s$ and $Q(s) \neq s$. If $Q = R_s$, it moves its anchor s. Otherwise $Q = R_t$, since $t \neq s$ and the other class has two elements, and it moves its anchor t. In either case D_3 makes Q a three-cycle. The only other three-cycle is P^{-1}, whose predecessor of s is different from t. This contradicts $Q(t) = s$. □

Lemma 2.3. *If f is a self-map of a set of size n and $f^k(x) = x$ for $k > 0$, the least positive return length of x is a divisor of k at most n.*

Proof. Let d be the least positive return length. Dividing k by d shows that the remainder is also a return length if it is positive, so it is zero. The first d iterates starting at x are distinct. Indeed, if $f^a(x) = f^b(x)$ for $0 \leq a < b < d$, applying f^{d-b} gives a positive return at $a + d - b < d$. Thus $d \leq n$. □

We can now prove necessity in Theorem 1.1. Suppose none of (1)–(3) holds. Under P_k apply Lemma 2.3 to every x in every column, and under D_k apply it to the anchor x in column R_x. At size at most two, all the resulting return lengths divide 2. At size three we have $6 \nmid k$, so the lengths divide a single common exponent in $\{2, 3\}$, since lengths 2 and 3 cannot both occur. In these cases Lemmas 2.1 and 2.2, with the reductions above, give every required conclusion. At size at least four, neither 2 nor 3 divides k and there is no divisor in $[5, n]$. The only possible return length is then 1, since $4 \mid k$ would imply $2 \mid k$. Under P_k this gives $xy = x$, which implies C_5. Under D_k it gives $xx = x$, which implies C_1. With B_5 this also gives C_2. Thus no countermodel exists in any of these cases.

3. Witnesses at every admissible size

We first construct small magmas satisfying P_k and B_5 in which $x(xx) \neq xx$ for a specified x. The reductions of Section 2 make each of these a countermodel to all the indicated implications. We compose permutations from right to left and use cycle notation in which $(0\,1\,2)$ sends 0 to 1.

3.1. **Three elements and exponent six.** On $\{0,1,2\}$ take $P = (0\,1)$ and $Q = (0\,1\,2)$. Put $R_1 = Q$ and $R_y = P$ for $y \neq 1$. Both P and Q send only 0 to 1. Thus the $\sim$-class of $R_z(y)$ is independent of z, giving B_5. We have $1 \cdot 1 = 2$ and $1 \cdot 2 = 0$. Both columns have order dividing 6. This table is isomorphic to the archived table in [6] via the relabeling $0 \mapsto 1$, $1 \mapsto 2$, $2 \mapsto 0$.

3.2. **Four elements and exponent two.** On $\mathbb{Z}/4\mathbb{Z}$ define $xy = -x + 2y + 1$. Each column is an involution, and the column R_y depends only on the parity of y. The parity of $yz = -y+2z+1$ depends only on y, so B_5 holds. Here $0 \cdot 0 = 1$ and $0 \cdot 1 = 3$. Exchanging the labels 1 and 2 gives the archived table in [7].

3.3. **Four elements and exponent three.** On $\{0,1,2,3\}$ take $P = (0\,1\,2)$ and $Q = (0\,1\,3)$. Put $R_1 = Q$ and $R_y = P$ for $y \neq 1$. Both columns send only 0 to 1, giving B_5 as in the three-element construction. We have $1 \cdot 1 = 3$ and $1 \cdot 3 = 2$. Both columns have order 3.

3.4. **A pair of conjugate cycles.** Fix $d \geq 5$ and work on $\mathbb{Z}/d\mathbb{Z}$. Let $P(x) = x + 1$, let $\tau = (0\,1)$, and put $Q = \tau P \tau$. With $A = \{0,1,2\}$, define

$$R_y = \begin{cases} P & \text{if } y \in A, \\ Q & \text{if } y \notin A. \end{cases}$$

The permutations agree except at $0, 1, d-1$, where their values are

y	0	1	$d-1$
$P(y)$	1	2	0
$Q(y)$	2	0	1

All displayed values belong to A. Thus $P(y)$ and $Q(y)$ are in the same $\sim$-class for every y. This is B_5. For $x = d-1$ we have $xx = Q(x) = 1$ and $x(xx) = P(x) = 0$. Since both columns are d-cycles, the witness satisfies P_d.

3.5. **Extension to larger sizes.** Let M be any of the witnesses just constructed, with exponent e, and choose $a \in M$. Adjoin new elements to obtain a set M'. Extend each old column by fixing every new element, and let each new column be the extension of R_a. The new columns have orders dividing e, so P_e holds on M'. For an old y, the products yz with new z equal ya, so all outputs yz have the required column R_{yy} after extension. For a new y, every yz equals y, which also gives B_5. The old table is a submagma, so its diagonal failure persists.

If $e \mid k$, the law P_e implies P_k, which in turn implies D_k. Using the size-three witness when (1) holds, the applicable size-four witness when (2) holds, and the d-cycle witness when (3) holds proves sufficiency. This completes the proof of Theorem 1.1.

4. Formal verification

The ancillary file `CountermodelSpectrum.lean` proves Theorem 1.1 in Lean 4 [3]. A multiplication table is represented as a function `Fin n → Fin n → Fin n`. The two biconditionals are

```
GDMD.periodicCountermodelSpectrum
GDMD.diagonalCountermodelSpectrum.
```

Both quantify over $k > 0$ and $n \in \mathbb{N}$. Their right-hand side, `GDMD.PeriodicThreshold`, expresses the disjunction in Theorem 1.1. The identity indices are zero-based in Lean. The diagonal theorem therefore uses side indices $3, 4$ and conclusion indices $0, 1$. The file also verifies the four arithmetic cases used in Corollary 1.2.

The orbit bound, the general cycle witness, and the enlargement construction are parametric proofs. For several of the small obstructions, the formal proof uses finite checks via `decide`. The checks for the full law at exponents two and three range over triples of involutions or triples of permutations of order dividing three, respectively. For the diagonal law at exponent three, the check ranges over columns that satisfy the return test at their anchors. These checks are evaluated by the kernel. To reproduce the verification with Lean 4.35.0, run the following in the directory containing the ancillary files.

```
lean -DwarningAsError=true CountermodelSpectrum.lean
```

Lean reports the same three axioms for both spectrum theorems.

`propext`, `Classical.choice`, `Quot.sound`.

5. Prior work and scope

5.1. **Cyclic groupoids and translation constructions.** Płonka's k-cyclic groupoids satisfy the full return law P_k together with idempotence, commuting right translations, and the reduction law C_5 [9, 11]. Divisibility governs the lattice of these varieties, and Romanowska and Roszkowska give a representation using abelian groups [10]. Agore, Chirvasitu, and Militaru [1, Theorem 3.24] characterize the right ideal-simple right Płonka magmas by a single full cycle used for every right translation. In the present construction two cycles give a common map from inputs to column classes, as required by B_5. This weaker condition leaves room for failure of C_5.

5.2. **Overlap with the Equational Theories Project.** The Equational Theories Project (ETP) studies implications between short single identities and the model spectra of individual laws [2]. Its archive already contains the size-three and affine size-four tables credited in Section 3 [6, 7]. There is also a direct overlap at exponent two. In the ETP numbering, the identity E1656, $x = (xy)((yx)z)$, implies E26 (P_2) and E4339 (B_5) [4]. Conversely, P_2 and B_5 give $R_{(yx)z} = R_{(yx)x} = R_y$, so $(xy)((yx)z) = (xy)y = x$. Hence ten of the written cases at $k = 2$ have the same premise theory as E1656. The ETP refutation data through order four already give minimum size four for E1656 against E307, E308, E325, E326, and E327, which are C_1 through C_5 in that order [5]. Le Floch studies a related classification of conjunctions of short identities in the associative setting [8].

5.3. **Contribution.** The contribution is the uniform countermodel spectrum for every positive exponent and both families of premises in Theorem 1.1. It includes the mixed-period threshold at multiples of six and the sharp, unbounded least-prime-divisor threshold when k is coprime to six. The reviewed works give precedents for the periodic laws, the use of cycles, and the small witnesses. The exact all-exponent spectrum for these implications was not found in those works.

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